

Part I

1. Free-body diagram
2. T
3. Torsional, Bending
4. Partially
5. F
6. 8
7. F
8. Improperly
9. T
10. Normal force, Shear force

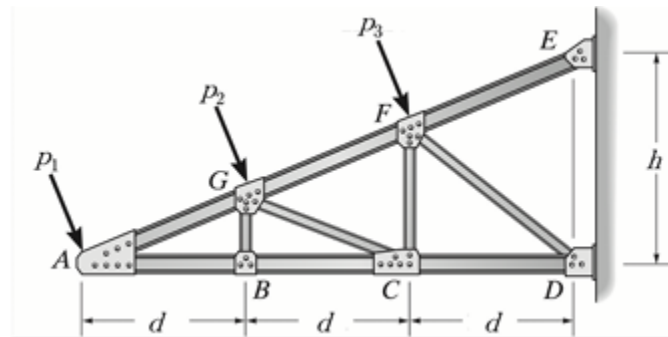
Scenario 1

11. True
A cantilever with a fixed support has three reaction components in 2D (A_x , A_y , M_z), but with only vertical load, $A_x = 0$ and we solve A_y and M_z directly using equilibrium. No extra supports; determinacy holds.
12. 2 kN and 2.2 kN·m
Sum of vertical forces: $A_y - P = 0 \Rightarrow A_y = 2 \text{ kN}$.
Sum of moments about support: $M_z - P \cdot L = 0 \Rightarrow M_z = 2 \text{ kN} \times 1.1 \text{ m} = 2.2 \text{ kN} \cdot \text{m}$.
13. 2.2 kN·m, 1.1 m
For a cantilever with end load, $M(x) = P \cdot x$ (measured from the free end). Max at $x = L$: $M_{\max} = 2 \times 1.1 = 2.2 \text{ kN} \cdot \text{m}$, occurring at the fixed support, which is 1.1 m from the free end.
14. 2 kN, 1.1 m
Shear is constant: $V(x) = P = 2 \text{ kN}$ along the span. It's commonly reported at the fixed end
15. 0
Shear is constant: $V(x) = P = 2 \text{ kN}$ along the span. It's commonly reported at the fixed end
16. 1
 $M(x) = P \cdot x$ is linear (degree 1)

Scenario 2

17. 0.5556 kN
18. 0.4444 kN
19. 0 kN·m
A pin cannot carry moment; internal bending moment at a free pin support is zero.
20. -0.4444 kN
21. 0 kN
Shear jumps by the reaction at the pin. Across the pin, internal shear to the right is zero because there's no span beyond the pin.
22. False
Position "0.9 from the right end" means $x = 0.1 \text{ m}$ from the right, $V(x)$ for $0.1 < x < 1$: $V(x) = R_r - w x = 0.5556 - x$. At $x = 0.9$, $V = -0.3444 \text{ kN}$, not zero.
23. 1
shear varies linearly: $V(x) = R_r - w x$.
24. 2
 $M'(x) = V(x) \Rightarrow M(x)$ is quadratic under constant w . Integrating $V(x)$ gives a degree-2 expression

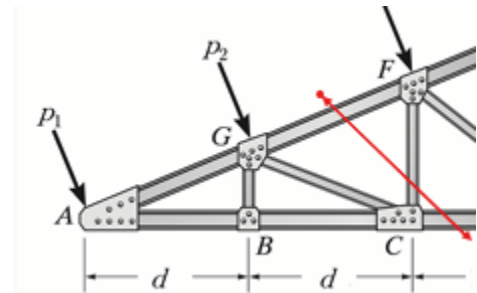
Part II. Where applicable, provide force diagrams and indicate the sense of member forces. Enclose final answers in boxes, with non-integer values rounded to three (3) decimal places.



Problem 1. Consider the truss subject to idealized loads as shown above. Assume that the members have negligible weights and that the joints behave as frictionless pins. The magnitudes of p_1 , p_2 , and p_3 are 5 kN, 6 kN, and 7 kN, respectively. Set the dimensional parameters d and h to 2 ft and 3 ft, respectively.

$$(1.788854382\text{ft})F_{FG} + (6\text{kN})\cos(63.43494882^\circ)(1\text{ft}) \\ = (6\text{kN})\sin(63.43494882^\circ)(2\text{ft}) + (5\text{kN})\sin(63.43494882^\circ)(4\text{ft})$$

- a) $F_{gf} = 14.500 \text{ kN (T)}$
 $F_{cd} = 17.889 \text{ kN (C)}$
 $F_{cf} = 3.354 \text{ kN (T)}$



$$\sum F_x = 3.130 - 12.97 + 0.894F_{EF} + 0.707F_{DF} = 0 \\ \sum F_y = -6.261 - 6.48 - 3.354 + 0.447F_{EF} - 0.707F_{DF} = 0$$

- b) $F_{EF} = 19.330 \text{ kN (T)}$
 $F_{DF} = 10.520 \text{ kN (C)}$
- c) Joint E is pin support and Joint D is the Roller.
- d) The zero-force members are AB, AG, BG, and BC.
- e) Reaction at E: Balances FEF directly.

$$R_E = F_{EF} = 19.330 \text{ kN}$$

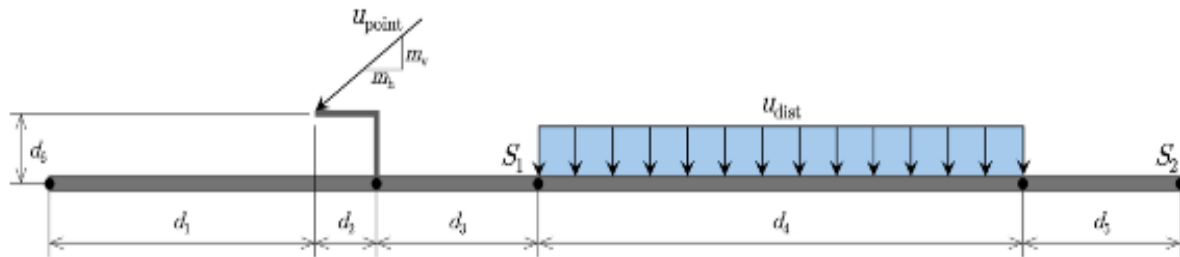
Reaction at D: Balances F_{CD} (Horizontal) and F_{DF} (Diagonal).

$$R_{Dx} = F_{CD} + F_{DF} \cos(45^\circ) = 17.889 + 10.520(0.707) = 10.45 \text{ kN}$$

$$R_{Dy} = -F_{DF} \sin(45^\circ) = -10.520(0.707) = -7.44 \text{ kN}$$

$$|R_D| = \sqrt{10.45^2 + 7.44^2} = 12.830 \text{ kN}$$

Problem 2. A custom beam design places supports at S1 and at S2, in anticipation of a concentrated load of upoint kips and a uniformly distributed load of intensity udist kip/ft. Your main task is to analyze how the internal loadings induced vary as a function of x, where x is the distance from the left end of the beam. Consider the case when upoint is 5 kips and udist is 2 kip/ft. Set the respective values of d1, d2, d3, d4, d5, d6 to 3.2, 0.8, 2.0, 6.0, 2.0, and 0.6 ft, and those of mv and mh to 3 and 4. For static determinacy, assume a pin at S1 and a roller at S2.



Note:

N= Normal Force

V= Shear Force

M_o = Bending Moment

$$\sum M_{5\text{kips}} = 5(3/5)(0.8) + 5(4/5)(0.6)$$

$$\sum M_{5\text{kips}} = 4.800 \text{ kip-ft}$$

A.) $\sum M_{S1} : 5(3/5)(2) + 4.8 + S_2(8) = 2(6)(3)$

$$S_2 = (36 - 4.8 - 6)/8$$

$$S_2 = 3.150 \text{ kip}$$

$$\sum F_x : S_{1x} = 5(4/5)$$

$$S_{1x} = 4.000 \text{ kip}$$

$$\sum F_y : S_{1y} = 5(4/5) + 2(6) - 3.15$$

$$S_{1y} = 11.850 \text{ kip}$$

B.) $\sum F_x : N = 5(4/5)$

$$S_{1x} = 4.000 \text{ kip}$$

C.) $\sum F_y : V + 5(3/5) = 0$

$$V = -3.000 \text{ kip}$$

D.) $\sum M_B : M_o + 4.8 = 0$

$$M_o = 4.800 \text{ kip-ft}$$

- $0 \leq x \leq 4$

$$N = 0$$

$$V = 0$$

$$M_o = 0$$

- $4 \leq x \leq 6$
 $N = 4.000 \text{ kip}$
 $V = -3.000 \text{ kip}$
 $\sum M_B: M_o = -4.8 - 5(3/5)(x-4)$
 $M_o = -4.8 - 3x + 4$
 $M_o = 7.2 - 3x$
- $6 \leq x \leq 12$
 $N = 0$
 $\sum F_y: V = -5(3/5) - 2(x-6) + 11.85$
 $V = 20.85 - 2x$
 $\sum M_B: M_o = -4.8 - 5(3/5)(x-4) - 2(x-6)[(x-6)/2] + 11.85(x-6)$
 $M_o = -4.8 - 3x + 12 - (x^2 - 12x + 36) + 11.85x - 71.1$
 $M_o = -x^2 + 20.85x - 99.9$
- $12 \leq x \leq 14$
 $N = 0$
 $\sum F_y: V = -5(3/5) - 2(6) + 11.85$
 $V = -3.15 \text{ kip}$
 $\sum M_B: M_o = -4.8 - 5(3/5)(x-4) - 2(6)(x-9) + 11.85(x-6)$
 $M_o = -4.8 - 3x + 12 - 12x + 108 + 11.85x - 71.1$
 $M_o = 44.1 - 3.15x$

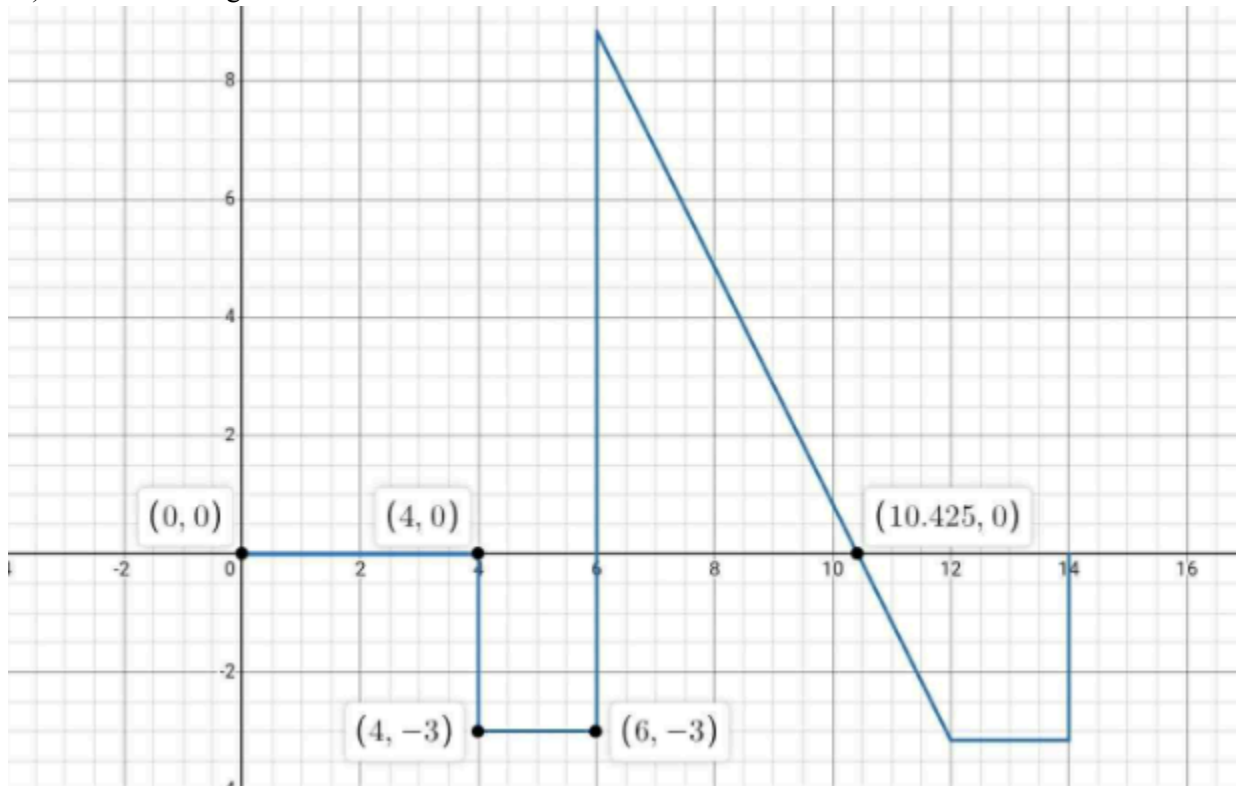
E.)

- $V;$
 0.000 , $0 \leq x \leq 4$
 -3.000 , $4 \leq x \leq 6$
 $20.85 - 2x$, $6 \leq x \leq 12$
 -3.150 , $12 \leq x \leq 14$

F.)

- $M_o;$
 0.000 , $0 \leq x \leq 4$
 $7.2 - 3x$, $4 \leq x \leq 6$
 $-x^2 + 20.85x - 99.9$, $6 \leq x \leq 12$
 $44.1 - 3.15x$, $12 \leq x \leq 14$

G.) Shear force diagram:



H.) Bending Moment Diagram:

